

# Linear Regression

Linear Regression is a **Supervised Machine Learning algorithm** used to predict **continuous numerical values** by modeling the relationship between independent variable(s) and dependent variable.

It assumes that the relationship between variables is **linear**.

General Idea:

We try to draw the best straight line that minimizes prediction error.

Used when:

- Output is continuous (price, salary, marks, temperature)
- We want interpretability
- We want to understand feature impact
- Data shows linear trend

Examples:

- House price prediction
- Salary prediction
- Sales forecasting
- Demand prediction

## Types of Linear Regression

### Simple Linear Regression

Used when there is **only one independent variable**

Model:

$$Y_p = mX + b$$

Where:

X = Independent variable

$Y_p$  = Predicted value

m = Slope

b = Intercept

Graph: Straight line in 2D

### Multiple Linear Regression

Used when there are **multiple independent variables**

Model:

$$Y_p = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_nX_n$$

Example:

$$\text{Salary} = b_0 + b_1(\text{Experience}) + b_2(\text{Education}) + b_3(\text{Skills})$$

Here:

$b_0$  = Intercept

$b_1, b_2, b_3$  = Coefficients

Graph: Hyperplane in multi-dimensional space

## Statistical Foundation

Mean of X:

$$\bar{X} = (\sum X) / n$$

Mean of Y:

$$\bar{Y} = (\sum Y) / n$$

Slope:

$$m = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}$$

Intercept:

$$b = \bar{Y} - m\bar{X}$$

## Intuition Behind Slope

Slope represents:

How much Y changes when X increases by 1 unit.

If  $m = 2$

Then when X increases by 1  $\rightarrow$  Y increases by 2

## Assumptions of Linear Regression

### 1. Linearity

Relationship between X and Y is linear

### 2. Independence

Observations are independent

### 3. Homoscedasticity

Constant variance of errors

### 4. No Multicollinearity (for multiple regression)

Independent variables should not be highly correlated

## 5. Normality of errors

# Cost Function

Linear Regression uses:

Mean Squared Error (MSE)

$$\text{MSE} = (1/n) \sum (Y - Y_p)^2$$

Goal:

Minimize MSE

Why square?

- Removes negative sign
- Penalizes large errors more

# Important Statistical Terms

Mean of X:

$$\bar{X} = (\sum X) / n$$

Mean of Y:

$$\bar{Y} = (\sum Y) / n$$

Slope (m):

$$m = \sum (X - \bar{X}) (Y - \bar{Y}) / \sum (X - \bar{X})^2$$

Intercept (b):

$$b = \bar{Y} - m\bar{X}$$

# Step-by-Step Calculation

## Step 1: Calculate Means

$$X = [1, 2, 3, 4, 5]$$

$$Y = [3, 4, 2, 4, 5]$$

$$n = 5$$

$$\sum X = 15$$

$$\Sigma Y = 18$$

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$$\bar{X} = 15 / 5 = 3$$

$$\bar{Y} = 18 / 5 = 3.6$$

## Step 2: Create Calculation Table

|   |   | $X - \bar{X}$ | $Y - \bar{Y}$ | $(X - \bar{X})(Y - \bar{Y})$ | $(X - \bar{X})^2$ |
|---|---|---------------|---------------|------------------------------|-------------------|
| 1 | 3 | -2            | -0.6          | 1.2                          | 4                 |
| 2 | 4 | -1            | 0.4           | -0.4                         | 1                 |
| 3 | 2 | 0             | -1.6          | 0                            | 0                 |
| 4 | 4 | 1             | 0.4           | 0.4                          | 1                 |
| 5 | 5 | 2             | 1.4           | 2.8                          | 4                 |

Now sum:

$$\Sigma (X - \bar{X})(Y - \bar{Y}) = 1.2 - 0.4 + 0 + 0.4 + 2.8 = 4$$

$$\Sigma (X - \bar{X})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

## Step 3: Calculate Slope

$$m = 4 / 10$$

$$m = 0.4$$

## Step 4: Calculate Intercept

$$b = \bar{Y} - m\bar{X}$$

$$b = 3.6 - (0.4 \times 3)$$

$$b = 3.6 - 1.2$$

$$b = 2.4$$

## Final Regression Equation

$$Y_p = 0.4X + 2.4$$

## Predicted Values (Yp)

| X | Actual Y | $Y_p = 0.4X + 2.4$ |
|---|----------|--------------------|
| 1 | 3        | 2.8                |

|   |   |     |
|---|---|-----|
| 2 | 4 | 3.2 |
| 3 | 2 | 3.6 |
| 4 | 4 | 4.0 |
| 5 | 5 | 4.4 |

## Residual (Error)

$$\text{Residual} = Y - Y_p$$

| X | Y | Yp  | Residual |
|---|---|-----|----------|
| 1 | 3 | 2.8 | 0.2      |
| 2 | 4 | 3.2 | 0.8      |
| 3 | 2 | 3.6 | -1.6     |
| 4 | 4 | 4.0 | 0        |
| 5 | 5 | 4.4 | 0.6      |

## Mean Squared Error (MSE)

Formula:

$$\text{MSE} = (1/n) \sum (Y - Y_p)^2$$

Squared residuals:

$$0.2^2 = 0.04$$

$$0.8^2 = 0.64$$

$$(-1.6)^2 = 2.56$$

$$0^2 = 0$$

$$0.6^2 = 0.36$$

$$\text{Sum} = 3.6$$

$$\text{MSE} = 3.6 / 5$$

$$\text{MSE} = 0.72$$

## R-Squared (R<sup>2</sup>)

R<sup>2</sup> measures how much variation in Y is explained by X.

Formula:

$$R^2 = 1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{tot}})$$

Where:

$$SS_{\text{res}} = \sum (Y - Y_p)^2$$

$$SS_{\text{tot}} = \sum (Y - \bar{Y})^2$$

## Step 1: Calculate $SS_{\text{tot}}$

| Y | Y - $\bar{Y}$ | Square |
|---|---------------|--------|
| 3 | -0.6          | 0.36   |
| 4 | 0.4           | 0.16   |
| 2 | -1.6          | 2.56   |
| 4 | 0.4           | 0.16   |
| 5 | 1.4           | 1.96   |

$$SS_{\text{tot}} = 5.2$$

## Step 2: $SS_{\text{res}}$

Already calculated = 3.6

## Step 3: Calculate $R^2$

$$R^2 = 1 - (3.6 / 5.2)$$

$$R^2 = 1 - 0.6923$$

$$R^2 = 0.3077$$

# Final $R^2 \approx 0.31$

Interpretation:

Model explains only 31% variance.

Weak linear relationship.

# Conceptual Understanding

$SS_{\text{tot}} \rightarrow$  Total variation in Y

$SS_{\text{res}} \rightarrow$  Unexplained variation

$$SS_{\text{exp}} = SS_{\text{tot}} - SS_{\text{res}}$$

Explained Variation:

$$5.2 - 3.6 = 1.6$$

$$1.6 / 5.2 = 0.31$$

Same  $R^2$ .

# Training Process Internally

1. Initialize  $m$  and  $b$  randomly
2. Calculate  $Y_p$
3. Compute error
4. Compute cost
5. Update  $m$  and  $b$
6. Repeat

This is called Optimization.